“Macro Components” in VBL++ for Condensing Long Chains of Components and Rapid Deck Creation

Aditi Thanekar
Directorate: NIF- COMP
Mentor: River Aden

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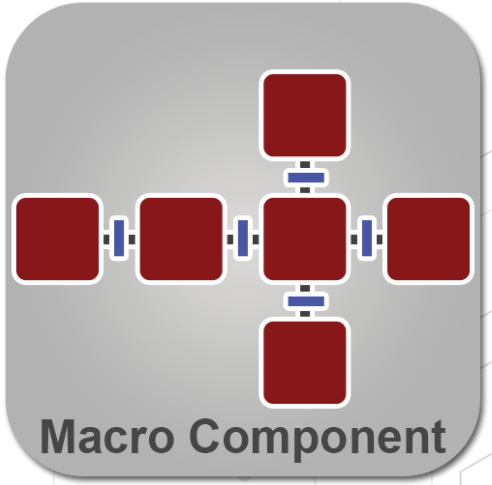
What is a Macro Component?

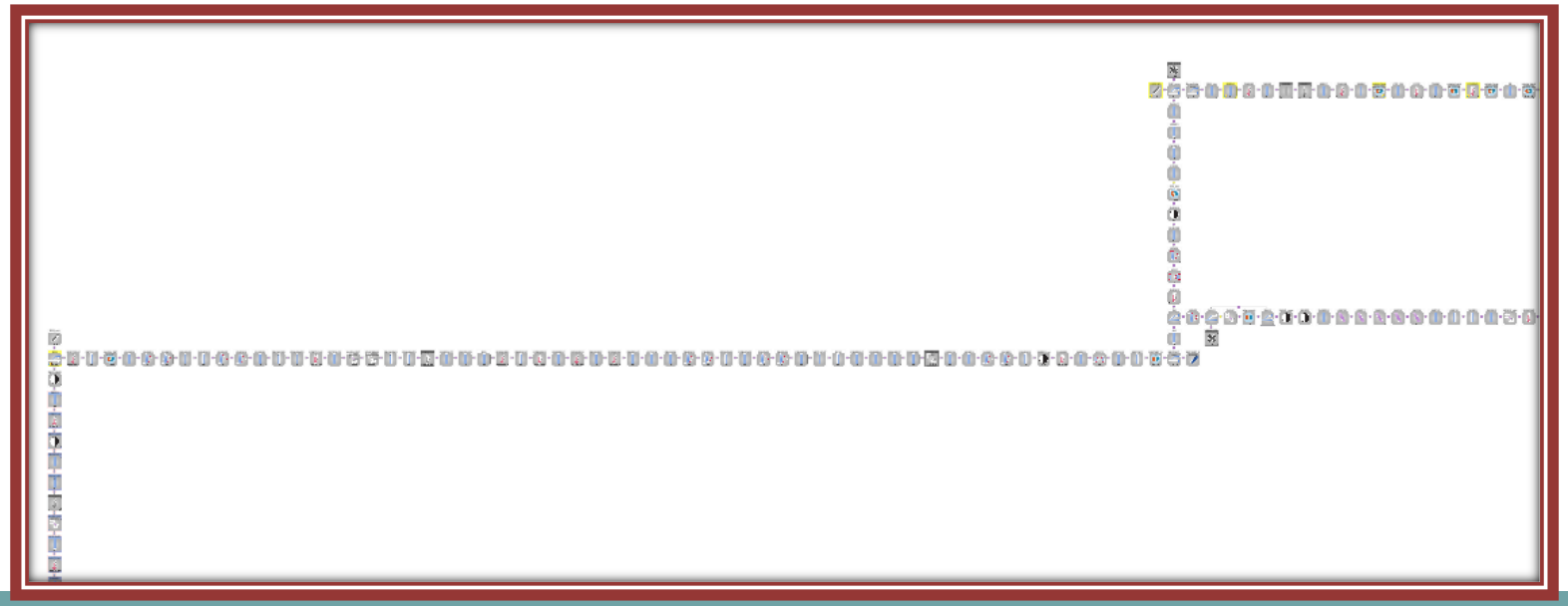
- A new VBL++ component in the component palette!

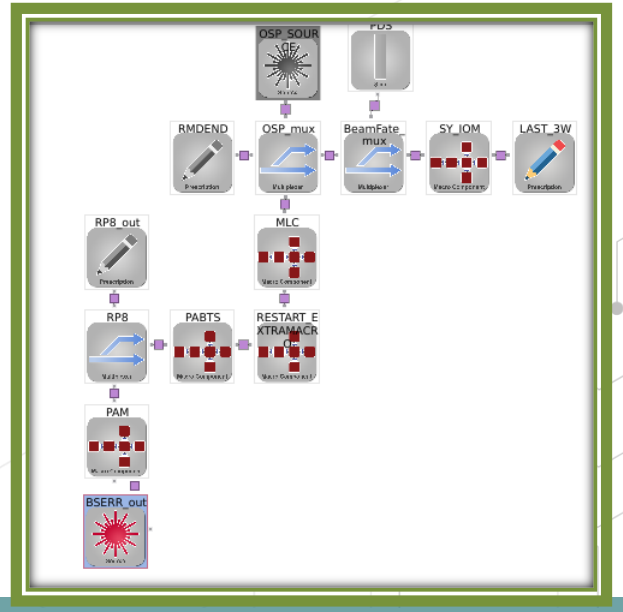
- Macro Components can contain chains of components,
- Helps to section off a set of components, visually pleasing
- “Borrows” components from a different .vpp file so they don’t have to be rewritten

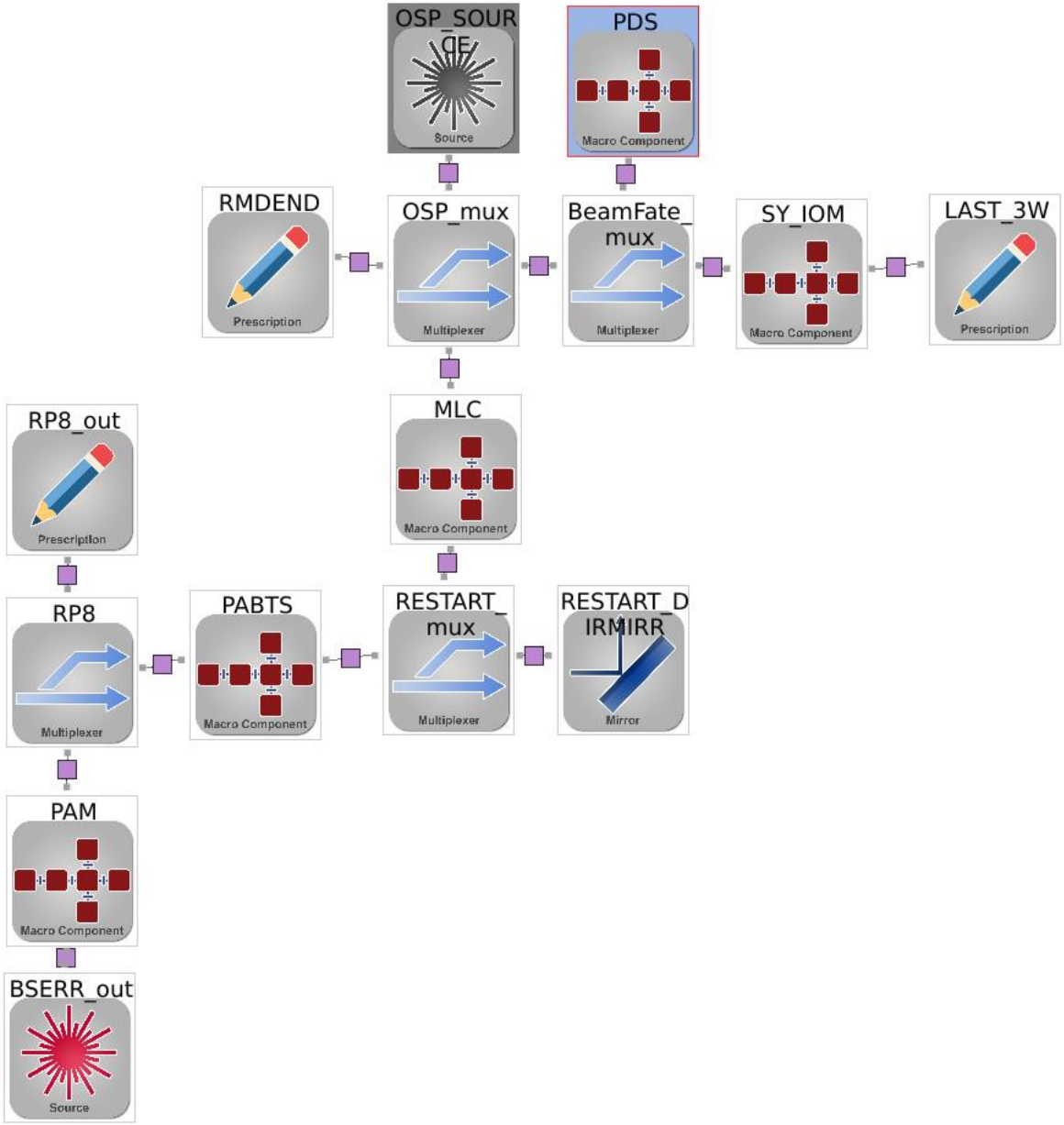


Why a Macro Component Matters to You!

- Quicker creation of the deck, reusing chains that are frequently used when creating a new deck from scratch
- Chunk sections into NIF beamline specified sections like MA, PA, etc.
- Mixing and matching sections of components from different files

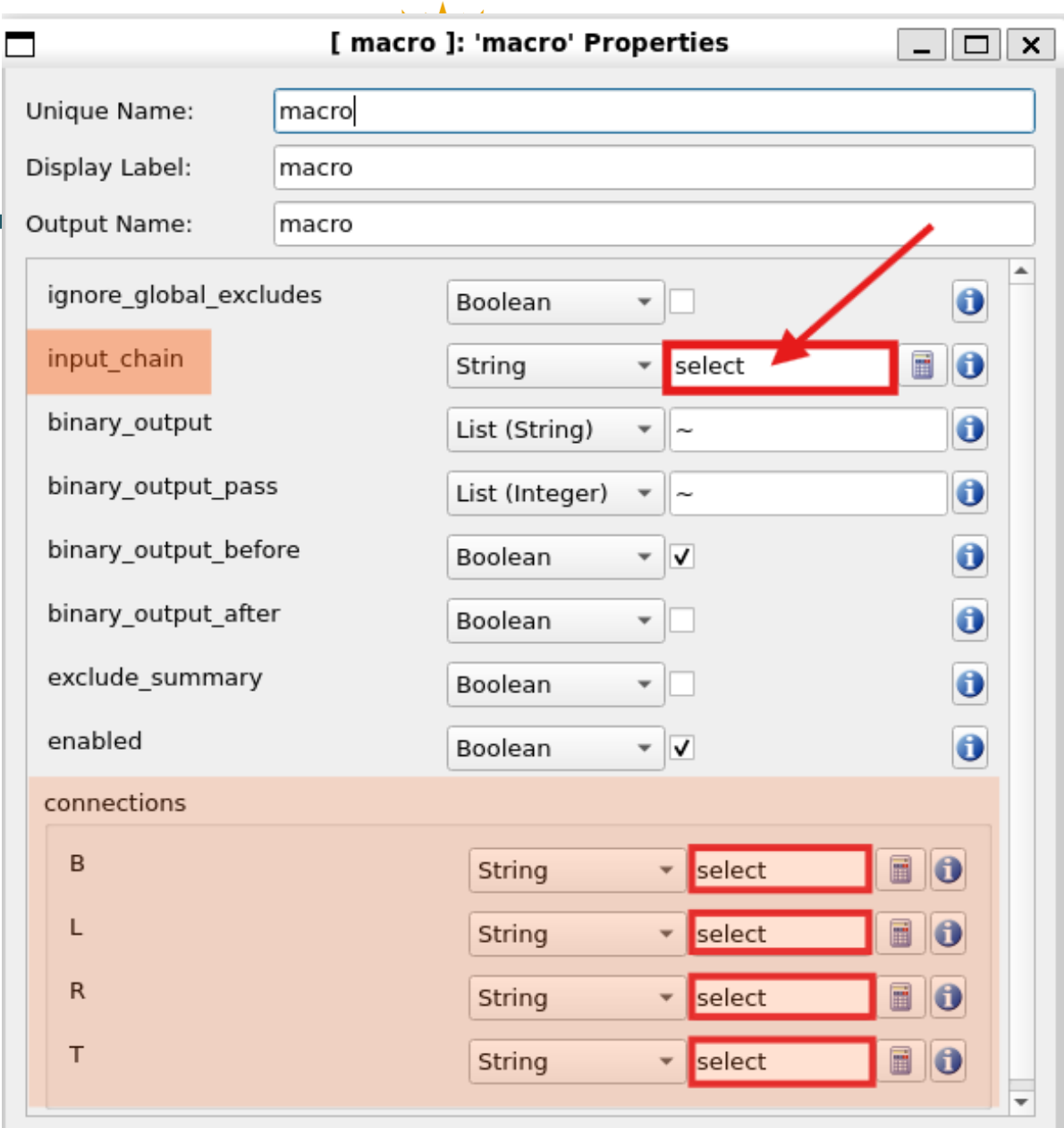





OSP_SOURCE
Source
PDS
Macro Component
RMDEND
Prescription
OSP_mux
Multiplexer
BeamFate_mux
Multiplexer
SY_IOM
Macro Component
LAST_3W
Prescription
MLC
Macro Component
RP8_out
Prescription
RP8
Multiplexer
PABTS
Macro Component
RESTART_mux
Multiplexer
RESTART_DIRMIRR
Mirror
PAM
Macro Component
BSERR_out
Source

Properties of a Macro Component

1. input_chain
2. connections(L, R, T, B)



Implementation that has been done (so far!)

- Macro Component is usable in the GUI, and CLI
- Able to contain Macro Components inside Macro Components
- Inherit calculation_parameters.input_dirs from an input_chain file
- Prefixes component names to avoid duplicate names
- Writes expanded version of components(without the Macro) to .verbose file
- clone() function created to clone a VBLSimulationObject (will help in future)

Limitations so Far (and bug fixes)

- Macro Component cannot be standing alone in a file without a source
- Macro Component is still being tested
- Not optimized for speed yet, if there are 2 Macro Components with the same input_chain, it loads the file in 2 separate times.
- If a mirror is not contained inside a Macro Component, and depends on Macro Components to determine its direction, it may ask to be rotated and not validate (so a mirror should also be in a Macro Component to fix this)

Implementation we want to add for the future-

In the future we want to:

- Open the Macro Component in a separate tab or window
- Click and expand a Macro Component individually
- Select components and have a button to rapid create Macro Component

Note: We are open to your suggestions for Phase 2!

Thank you all for listening, and a special thank you to the VBL++/LPOM team for a great summer! ☺

Questions?

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